

# Wenbin Wang, Master.

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## Research Interest

My research lies at the intersection of machine learning, statistics, and optimization, with a focus on the theoretical foundations of large language models and the development of AI algorithms for large-scale and multi-agent decision-making. I am particularly interested in understanding the mechanisms of large language models, including their training and prompting processes, and leveraging these insights for applications like decision-making. Recently, my work has focused on reinforcement learning from human feedback (RLHF) and the mathematical foundations of reinforcement learning.

## Education

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|-------------------|---------------------------------------------------|----------------------------------------|
| 09/2023 – 06/2026 | 📖 <b>M.Eng. Engineering</b>                       | <b>Computer Science and Technology</b> |
|                   | Advisor: Prof. Ziping Zhao & Prof. Xavier Lagorce | ShanghaiTech University                |
| 09/2018 – 06/2022 | 📖 <b>B.Econ. Economics</b>                        | <b>Economic Statistics</b>             |
|                   | Advisor: Prof. Zeyu Zhang                         | Huazhong Agricultural University       |

## Professional Experience

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|-------------------|-----------------------------------|-----------------------------------------------|
| 06/2025 – 09/2025 | 📖 Research Intern (remote)        | Massachusetts Institute of Technology         |
|                   | Worked on ensemble learning.      | Advised by Prof. Jinhua Zhao & Dr. Xinyu Chen |
| 01/2025 – 04/2025 | 📖 Research Intern (remote)        | Carnegie Mellon University                    |
|                   | Worked on reinforcement learning. | Advised by Dr. He Wang                        |
| 07/2024 – 09/2024 | 📖 Research Intern (remote)        | Duke University                               |
|                   | Worked on federated unlearning.   | Advised by Prof. Minghong Fang                |
| 04/2023 – 08/2023 | 📖 Visiting Student                | ShanghaiTech University                       |
|                   | Worked on compressed sensing.     | Advised by Prof. Ziping Zhao                  |
| 09/2022 – 03/2023 | 📖 Research Intern (remote)        | KU Leuven University                          |
|                   | Worked on 3D Reconstruction.      | Advised by Dr. Xueyang Kang                   |

## Research Publications

### Conference Proceedings

- 1 **W. Wang**, X. Ping, C. Cheng, and Z. Zhao, “Noisy bilinear low-rank matrix sketching”, in *Proc. IEEE Information Theory Workshop (ITW)*, 2025.
- 2 **W. Wang**, Y. Shi, and Z. Zhao, “High-dimensional tensor regression with oracle properties”, in *Proc. The International Conference on Machine Learning (ICML)*, 2025. [\[Paper\]](#).
- 3 **W. Wang** and Z. Zhao, “Achieving oracle rate for large covariance matrix estimation from quadratic measurements”, in *Proc. IEEE International Symposium on Information Theory (ISIT)*, 2025.
- 4 **W. Wang** and Z. Zhao, “Optimal compressive covariance sketching via rank-one sampling”, in *Proc. The International Conference on Sampling Theory and Applications (SampTA)*, 2025. [\[Paper\]](#).
- 5 **W. Wang\***, Q. Ma\*, Z. Zhang, Y. Liu, Z. Liu, and M. Fang, “Poisoning attacks and defenses to federated unlearning”, in *Proc. The Web Conference (WWW)*, 2025. [\[Paper\]](#).

### Preprint & Manuscript

- 1 **W. Wang**, Z. Jing, Y. Shi, and Z. Zhao, "Low-rank covariance matrix estimation from quadratic measurements", Submitted to Neural Information Processing Systems (NeurIPS) 2025 (Under review).
- 2 **W. Wang**, J. Zhang, Y. Yan, J. Zeng, and Z. Zhao, "Achieving oracle statistical property for low-rank matrix recovery with application to image denoising", Submitted to Neural Information Processing Systems (NeurIPS) 2025 (Under review).
- 3 **W. Wang** and Z. Zhao, "Accelerating graph analytics via bilinear sketching", Manuscript in preparation for submission to IEEE Trans. Signal Process.
- 4 **W. Wang** and Z. Zhao, "Compressive covariance matrix sensing via quadratic sampling based on nonconvex learning", Manuscript in preparation for submission to IEEE Trans. Inf. Theory.
- 5 **W. Wang** and Z. Zhao, "Large low-rank covariance matrix estimation with oracle rate via majorization-minimization", Manuscript in preparation for submission to IEEE Trans. Signal Process.
- 6 **W. Wang\*** and H. Wang\*, "Robust constrained reinforcement learning with linear function approximation", Submitted to Neural Information Processing Systems (NeurIPS) 2025 (Under review).
- 7 **W. Wang\***, Q. Ma\*, Z. Zhang, Y. Liu, Z. Liu, and M. Fang, "Federated unlearning under attack: Poisoning threats and defensive strategies", Submitted to IEEE Trans. Dependable Secure Comput. (Under review), 2025.

## Skills

|                 |                                                                                    |
|-----------------|------------------------------------------------------------------------------------|
| Languages       | Strong reading, writing and speaking competencies for English, Mandarin Chinese.   |
| Coding          | C/C++, Python, R, $\text{\LaTeX}$ , Tensorflow, Pytorch, MATLAB.                   |
| Developer Tools | Git, Docker, Visual Studio Code, Jupyter Notebook, Linux Shell.                    |
| Misc.           | Academic research, teaching, training, $\text{\LaTeX}$ typesetting and publishing. |

## Miscellaneous Experience

### Awards and Achievements

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025 | <ul style="list-style-type: none"> <li>IEEE ITW Travel Grant Award, ITW 2025 Travel Grant Selection Committee</li> <li>IEEE ISIT Travel Grant Award, ISIT 2025 Travel Grant Selection Committee</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 2022 | <ul style="list-style-type: none"> <li>Outstanding Graduate, College of Economics and Management, Huazhong Agricultural University.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 2020 | <ul style="list-style-type: none"> <li>First Prize, Hubei Provincial Division, Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM).</li> <li>Practical Innovation Award, College of Economics and Management, Huazhong Agricultural University.</li> <li>Academic Progress Award, College of Economics and Management, Huazhong Agricultural University.</li> <li>Second Prize, 10th MathorCup College Mathematical Modeling Challenge.</li> <li>First Prize, 17th May 1st Mathematical Contest in Modeling.</li> <li>Second Prize, 13th "Certification Cup" National Undergraduate Mathematical Modeling Challenge.</li> </ul> |
| 2019 | <ul style="list-style-type: none"> <li>Merit Student Award, Huazhong Agricultural University.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

## Teaching

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|-------------|--------------------------------------------------|
| Spring 2025 | Teaching Assistant. SI231B: Matrix Computations. |
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## Miscellaneous Experience (continued)

Fall 2024        **Teaching Assistant.** CS182: Introduction to Machine Learning.

## Professional Activity

2025        **Conference Reviewer.** ICASSP2025.

## Amateur Experience

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|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| 2023    |  Quantitative Algorithm Developer (online)<br>Worked on statistically robust backtesting methodologies. | WhaleQuant(Github 1.2K ★)  |
| 04/2022 |  Data Analyst<br>Worked on data collection, cleaning and processing.                                    | China Securities Co., Ltd. |